
When to Stop: Operational Reliability and Enforceable Accountability for Agentic AI in Government Deployment

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Abstract

Agentic AI systems can become operationally unreliable before they produce a visible hallucination. We present Local Threshold Evaluation (LTE), a pre-deployment scaffold that detects reliability precursors, expansion drift, latency cliffs, context decay, repetition, and persistent contract failure, and maps them to intervention states (continue, retry, repair, escalate, abort). Across seven language models and six runs per model, every model landed in the same intervention regime on every run. The central finding is a divergence between static probe performance and rolling stress reliability: Mistral-7B matched Claude Haiku 4.5 on static contracts yet triggered an abort under accumulated context, while higher-failure models remained stable. The scaffold emits a structured audit artifact: a retained record of what the deploying organisation had reason to know before deployment.

1. Introduction

Consider the following scenario, which is not speculative. A government department deploys an AI-assisted pipeline to support an analyst team processing benefit eligibility determinations. The system is capable: it passed all relevant assessments at procurement, the model card is in order, and the deployment team ran it through a set of demonstration cases before going live. Six weeks in, the analysts have become reliant on the system’s summaries. What none of them know, because no instrument was in place to surface it, is that as the context window fills over the course of a long session, the model has begun silently misapplying instructions. It is generating authoritative-looking outputs that contradict the current policy. The text is coherent. The determinations are wrong.

The harm accumulates quietly. When it surfaces, the question of who is responsible is not straightforward: the analysts acted on the outputs they were given, the deployment team followed the documented procurement process, and

the model itself performed within the stated parameters of its capability assessments. You cannot hold a deploying organisation responsible for ignoring a warning it was never equipped to receive. An organisation that ran a pre-deployment reliability evaluation, received an explicit signal, and continued without establishing oversight has, by contrast, made a traceable decision. What the technical instrument produces is not a liability determination but the evidentiary basis for one.

This paper describes one such instrument. The contributions are: (i) empirical evidence that operational unreliability has detectable precursors before visible hallucination; (ii) a reproducible scaffold specification with four metrics, six trigger conditions, and an intervention logic mapping to five machine-readable recommendations; (iii) a seven-model study isolating static-probe vs rolling-stress divergence as the central empirical wedge; and (iv) a structured audit artifact that gives the foreseeability condition for liability a technical substrate.

2. Related Work

Technical AI governance and operationalisation. [Reuel et al. \(2024\)](#) surveyed AI policy documents from the EU, US, and China and found systematic disconnects between the capabilities governance actions assume and the technical tools that exist, including reliable evaluation methodologies and deployment verification tools. Their five-capacity taxonomy (assessment, access, verification, security, operationalisation) places this paper in the operationalisation strand. [Barrett et al. \(2023\)](#) adapt the NIST AI RMF for general-purpose AI systems but direct their profile at developers, not deploying organisations. [Brundage et al. \(2020\)](#) proposed audit trails, third-party evaluations, and structured red-team exercises as mechanisms for verifiable claims; most remain underspecified six years on. [Harack et al. \(2025\)](#) extend the verification agenda to international governance; the deployment-level evidence their framework would aggregate has no standard form, and LTE is a proposal for that form. [Marino et al. \(2025\)](#) apply an analogous argument to machine unlearning for EU AI Act compliance: identify the governance requirement, diagnose the gap, propose a direction.

Evaluation as a governance problem. Bommasani et al. (2023) established HELM across 30 models. The Foundation Model Transparency Index (Wan et al., 2025) finds average transparency fell from 58/100 in 2024 to 40/100 in 2025, with providers particularly opaque about post-deployment monitoring. Eriksson et al. (2025), in a meta-review of approximately 100 studies on AI evaluation shortcomings, identify construct validity failures, contamination, and gaming; the disproportionate-trust concern is illustrated precisely by the Mistral result below. Ibrahim et al. (2025) argue that single-prompt static evaluations miss harms arising from sustained interaction, which is what LTE’s rolling stress probes implement. Perez et al. (2023) show that automated, targeted behavioural testing surfaces failure modes that aggregate capability scores average away. Casper et al. (2024) prove formally that black-box access cannot produce behavioural guarantees from finite queries without additional assumptions; LTE’s probe contracts are precisely the additional structure the impossibility result identifies as necessary. Longpre et al. (2024) argue for a legal and technical safe harbour for evaluation research; the present contribution is infrastructure that makes structured auditing tractable within existing constraints.

Long-context degradation. Liu et al. (2024) showed that performance drops exceeding 30 percentage points on multi-document QA when relevant content is placed mid-context rather than at the boundaries. Du et al. (2025) show that performance degrades 13.9–85% across five models even with perfect retrieval, ruling out retrieval failure as the sole mechanism and establishing context decay as a structural feature of current architectures.

Governing agentic systems. Shavit et al. (2023) propose seven practices for governing agentic AI, including attributability and automatic monitoring; LTE is a technical implementation of both. Kolt (2025) identifies information asymmetry, discretionary authority, and loyalty as core agent-governance problems and proposes inclusivity, visibility, and liability in response; the visibility principle is what the audit trail produces. Kolt et al. (2026) argue that governance frameworks must specify the technical means by which legal concepts such as foreseeability and duty of care are instantiated in deployment decisions. Hammond et al. (2025) characterise multi-agent risk; LTE addresses the prior question of individual-model reliability that multi-agent reasoning presupposes.

Responsibility and accountability. Matthias (2004) identified the responsibility gap: when a machine’s behaviour cannot be predicted, neither designer nor operator can be held responsible. Danaher (2016) extends the analysis to four distinct gaps, including public accountability, which a retained pre-deployment audit trail closes where standard capability documentation cannot. Santoni de Sio & van den

Hoven (2018) ground accountability in tracking and tracing conditions; LTE’s audit trail is an implementation of the tracing condition. Kiener (2025) reframes the problem as responsibility abundance rather than a vacuum, requiring advance specification of which actor is accountable under which conditions; that is the function of the responsibility map in Section 7.

3. The Accountability Gap

3.1. Ex Ante Governance and Its Limits

The past three years have produced a substantial body of AI governance infrastructure. The EU AI Act introduced mandatory risk classification and documentation requirements for high-risk AI systems. The NIST AI Risk Management Framework provided voluntary guidance on risk identification and mitigation. The UK and US established AI safety institutes with mandates for model evaluation and pre-deployment testing. Sector-specific frameworks have begun to appear in healthcare, finance, and national security contexts. Across these instruments, the orientation is consistent: identify and mitigate risks before deployment.

This orientation is correct as far as it goes. Yet the ex ante frame has a structural limitation that becomes more acute as systems become more agentic. It evaluates a model’s behaviour under assessment conditions; it does not answer how the model will behave as context accumulates across a real multi-turn session, when instructions from many turns ago compete with current ones, or when the pipeline is handling its four hundredth interaction of the day rather than its first. The assessment condition and the deployment condition are not the same, and the gap between them is precisely where the failure modes that matter most for accountability tend to occur.

The ex post consequence is that liability is difficult to assign. The cases that surface are salient enough to trigger complaints or investigations; the cases that do not surface, the gradual, compounding errors in automated pipelines that produce no single visible incident, remain invisible to the institutions designed to catch them.

3.2. The Moral Hazard and the Responsibilisation Problem

The absence of systematic ex post accountability produces a predictable moral hazard. Responsibility is pushed downward onto individual users and frontline officials, even in cases where those actors neither designed nor meaningfully controlled the systems they are required to operate. A government analyst operating an AI-assisted pipeline is not equipped to detect the failure modes described in Section 4; that is a structural feature of the deployment, not a criticism of the analyst. The same applies up the chain: a director who

authorised the deployment made an organisational decision with systemic consequences, but without a pre-deployment reliability record, that decision leaves no traceable accountability anchor.

Soft governance mechanisms are insufficient to correct this dynamic for systems with high autonomy and high impact. What is needed is a mechanism that makes responsibility explicit at the point of deployment: one that generates a record establishing what the deploying organisation knew, what signals it received, and what decisions it made in response.

3.3. What Enforcement Requires, Technically

For ex post accountability to be enforceable rather than merely asserted, three conditions must hold: the deploying organisation must have had access to a reliability instrument capable of detecting the relevant failure modes before harm occurred; that instrument must have generated a structured output retained as a record; and the output must be interpretable by both technical and non-technical actors so that decisions made in response to or in disregard of those signals are attributable. The central governance argument of this paper is that these conditions are technically achievable at reasonable cost. The sections that follow describe one implementation.

4. Failure Is Invisible Without Instrumentation

The dominant image of AI failure is a hallucination. That case exists, yet the more common shape of operational failure in agentic workflows is gradual and instrumentally invisible: a deterioration across multiple dimensions simultaneously, well before any individual output would be identified as wrong by a human reviewer. Five such modes appear across the probe battery. *Over-expansion* produces more output than the task requires; across hundreds of calls per session, this consumes latency and token budgets and eventually breaks downstream parsers. *Near-cap pressure* pushes against maximum output limits so that structured formatting breaks at the tail (a JSON object cut off before its closing brace cascades into pipeline failure). *Context decay* is the most dangerous mode for government deployment: as conversation accumulates, the model silently misapplies or ignores instructions while outputs remain coherent, so a benefits eligibility pipeline can produce authoritative-looking determinations that contradict the current policy with no visible signal. *Latency cliffs* appear when response time under growing context exceeds what any downstream system can tolerate; nine seconds per step in an automated pipeline breaks the system’s capacity to function, regardless of output quality. *Repetition and persistent failure* appears when the model recycles prior phrasing or fails three consecutive steps under the same conditions: one bad step is noise, three is a behavioural state.

All five modes are gradual, hard to notice without purpose-built detection, and easy for a deploying organisation to disclaim responsibility for, precisely because no instrument was in place to surface the signal. The accountability implication is direct: if no instrument exists to detect failure before it produces harm, the foreseeability condition for liability cannot be established.

5. The Accountability Signal: LTE

LTE (Local Threshold Evaluation) is a reliability and intervention pipeline designed to detect the failure modes described above and emit an explicit, machine-readable accountability signal. The framework runs a compact battery of behavioural probes against a model, measures its outputs along several operational dimensions, and maps the results to one of five intervention states: *continue*, *retry*, *repair*, *escalate*, or *abort*.

5.1. Detecting Precursors, Not Hallucinations

The central claim of LTE is that operational unreliability has detectable precursors. A model can remain syntactically fluent, semantically plausible, and superficially compliant with a task specification while crossing measurable operational thresholds that make it unsuitable for unattended use in a bounded workflow. The scaffold does not attempt to detect hallucinations, factual errors, or misaligned behaviour in the classical sense. It detects the prior condition: the set of behavioural signatures that appear once a model has entered an operational regime where continued reliance is no longer justified.

This framing distinguishes LTE from adjacent approaches. It is not a hallucination detector (metrics are orthogonal to factual accuracy), not a capability evaluation (probes are deliberately narrow), and not an output validator (most of the signal arrives before any individual output would fail a schema check).

The governance relevance follows directly. An instrument that detects hallucinations after the fact is useful for forensic review. An instrument that detects the conditions under which a model is about to become unreliable is useful for the prior decision: whether to deploy at all, whether to route through human review, whether to treat the audit trail as a standing part of the deployment record.

5.2. Measurement Layer

The measurement layer tracks four quantities per generation: **Expansion Ratio** (output tokens over input tokens; detects over-expansion), **Length Overrun Rate** (a binary flag when output reaches 95% of the generation cap; detects near-cap pressure), **Runaway Continuation Score** (4-gram repetition fraction; detects repetition loops), and **gated Latency**

(wall time per generation; gated by input-token threshold to exclude warm-up steps). Exact formulas, trigger thresholds, and the intervention decision logic are in Appendices B and C; every value used in this paper is drawn directly from the scaffold’s committed configuration files.

These are operational health signals rather than capability measures. A model can produce semantically coherent output while all four metrics are trending in the wrong direction, which is precisely the case in which governance frameworks relying on capability benchmarks alone will miss the failure. Six trigger families convert the per-generation metrics into scaffold-level states: `over_expansion`, `near_cap_pressure`, `latency_cliff`, `repetition_loop`, `context_decay`, and `persistent_failure`. A trigger fires when any of its stated conditions holds; the aggregate trigger vector maps deterministically to one of five intervention recommendations (`continue`, `retry`, `repair`, `escalate`, `abort`) via an ordered rule set whose first rule sends any run with three-plus firing triggers or persistent failure to `abort`.

5.3. Probe Battery

The probe battery is organised into four families, each targeting a distinct failure mode that occurs in government-relevant agentic workflows.

Bounded-output probes test whether the model can stay within strict communication constraints on short operational tasks: a one-sentence incident summary, a two-bullet policy comparison, a concise handoff note. These appear trivial and fail surprisingly often under automation. Failure here means a downstream system received more data than it can process.

Structured-output probes test whether the model produces machine-parseable output under explicit schema constraints (JSON extraction, CSV emission, line-oriented action lists). Failures here are directly responsible for pipeline breakdowns and are recoverable only if the model can be prompted into producing the correct format. The probe tests not whether the model *can* produce structured output in principle, but whether it does so consistently under the conditions of actual deployment.

State-integrity probes test whether the model preserves stable state across context: whether it applies only current instructions rather than stale ones, whether it correctly carries over unchanged fields when updating a running record. This probe family is most directly relevant to agentic workflows where the model maintains state across many turns. Context decay often looks like normal output until something downstream depends on a field that was silently overwritten several turns ago.

Rolling stress probes run a fixed-output task repeatedly with an ever-growing context window, inserting prior outputs as read-only reference. The purpose is to find the point at which continued model use becomes operationally unjustifiable (through latency, repetition, or persistent contract failure) under conditions that approximate a real multi-turn agentic session. This is the probe family that identifies latency cliffs and persistent failure, and it is the family most likely to diverge from standard benchmark assessment.

5.4. Contracts and Failure Classification

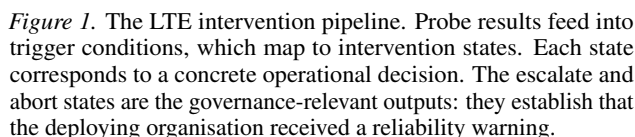
Each probe carries an embedded **contract** specifying exactly what a passing response looks like: required key sets, exact bullet counts, sentence limits, invariant fields that must be preserved. Contracts make the pass/fail determination mechanical rather than inferential. Appendix A gives all fourteen probes with their contracts; representative examples include a one-sentence incident brief capped at 22 words (recoverable), a JSON extraction with a required five-key schema (recoverable if shape is close, unrecoverable if the model refuses structure), and a state-reconciliation task where unchanged fields must be bit-identical to the input (unrecoverable: if the model silently overwrites stable fields, retrying reproduces the error).

The recoverable/unrecoverable classification is the governance-relevant distinction. Recoverable failures route to `retry` or `repair`. Unrecoverable failures contribute toward `escalate` or `abort`, because the model has entered a state that additional prompting will not resolve. Treating all failures as equivalent, as aggregate benchmark scores implicitly do, obscures the information that determines accountability.

5.5. Intervention States and the Audit Trail

The five intervention states each map to a concrete downstream decision: `continue` (signals are clean; proceed), `retry` (isolated recoverable failure; try once more with tighter constraints), `repair` (output shape is wrong but content is present; apply a deterministic correction), `escalate` (the model is still producing output but cannot be trusted without human oversight), and `abort` (continued use is unjustifiable; stop the pipeline). Figure 1 illustrates the decision logic connecting probe results, trigger conditions, and intervention states.

The governance-critical component is the **audit trail**. Every scaffold run produces a structured summary artifact containing trigger states, failure classifications, probe-level results, and the final intervention recommendation, retained as part of the deployment record. An organisation that ran this evaluation, received an `escalate` signal, and proceeded to deploy without routing outputs to human review has made a traceable decision. The framework does not determine

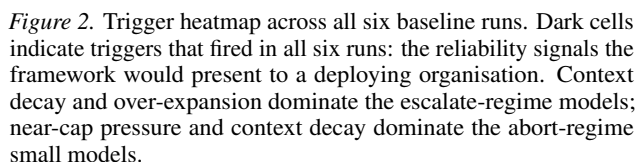


6. Results: Different Models, Different Ways of Failing

6.1. Regime Stability

Every model landed in the same intervention regime across all six baseline runs. A framework that returned different verdicts on re-run would be useless as an audit instrument: an organisation that received an `abort` signal on Monday and a `continue` on Tuesday would have no reliable basis for a deployment decision, and no accountability record that could support a liability determination. The consistency of classification across temperature and token-budget variations (the regime results were preserved at temperature 0.5 and at both 128- and 256-token max-output budgets) supports the interpretation that the scaffold is detecting something persistent about each model’s operational character rather than an artifact of a specific prompt or decoding configuration.

Two local models, Phi-4-mini and Meta-Llama, consistently fell into an `escalate` regime. Both showed the same dominant failure pattern: context decay combined with over-expansion. Phi-4-mini averaged seven probe contract failures per run; Meta-Llama averaged nine. Both remained operationally functional (they did not collapse under stress), but an `escalate` signal means human oversight is required before any output from these models is acted on in



6.2. The Static-Probe vs Stress Divergence: Mistral

The most governance-consequential result is Mistral, and it deserves careful treatment. Mistral-7B-Instruct-v0.3 averaged only four probe contract failures per run, fewer than either of the `escalate`-regime local models, and matching Claude Haiku 4.5 in aggregate. A procurement process ranking these models on probe contract performance alone would rank Mistral ahead of both Phi-4-mini and Meta-Llama.

Figure 3 shows what happened under stress.



As context accumulated, Mistral’s response latency climbed linearly: 3.9 seconds at step 0 (177 input tokens), 6.9 seconds at step 10 (1094 input tokens), and 8.3 seconds at step 14 (1442 input tokens). The first gated latency failure ap-

peared at step 15 (8.7 seconds, 1529 input tokens), and by step 17 it had logged three consecutive latency failures, triggering persistent failure and the `abort` recommendation. Nine seconds per response may not sound large in isolation. In an automated pipeline making dozens of sequential calls per session, it means the system either stalls on each step or users abandon it before completion. Latency at that level does more than degrade the workflow. It breaks it. Appendix E gives the full per-step trajectory.

Haiku ran all 22 stress steps without crossing the latency threshold, landing in an `escalate` regime. The two models make a precise contrast at identical probe contract failure counts: Mistral scored the same on the probe contracts but became operationally unusable at scale; Haiku stayed operationally alive but cannot be trusted without oversight because context decay fired in every run. A standard capability score would treat them as equivalent. The scaffold separates them correctly, in a direction that directly changes the procurement decision.

The remaining local models, Phi-3-mini and SmoLLM, followed a different `abort` pathway: over-expansion, near-cap pressure, context decay, and persistent failure firing together, with stress failure beginning at step 2. SmoLLM is the clearest negative case in the experiment: the highest total failures, the highest proportion of unrecoverable failures of any model tested, and near-immediate stress collapse that makes it unsuitable for any pipeline requiring more than a single-turn interaction.

6.3. Recoverable Versus Unrecoverable: The Severity Distinction

SmoLLM failing six of its twelve probe failures unrecoverably is a different situation from Phi-4-mini failing three of its seven. Aggregate failure counts treat both as equivalent; they are not. A model whose failures are predominantly recoverable can be managed through constrained retry logic and deterministic repair. A model whose failures are predominantly unrecoverable has entered states that additional prompting will not resolve, and deploying such a model in an automated pipeline is a decision the audit trail will document. Figure 4 gives the breakdown.

6.4. Scaffold Transfer to API Backends

Both API models completed the full 22-step stress battery, a baseline result that the local-model cohort could not match. GPT-4o mini averaged 3.5 probe contract failures per run (the lowest of any model tested) and landed in a `retry` regime: over-expansion fired in every run, but the failures were recoverable and context decay did not fire. Claude Haiku 4.5 averaged 4.0 failures, marginally higher, and landed in an `escalate` regime. The deciding trigger was context decay, which fired in all six Haiku runs and in none

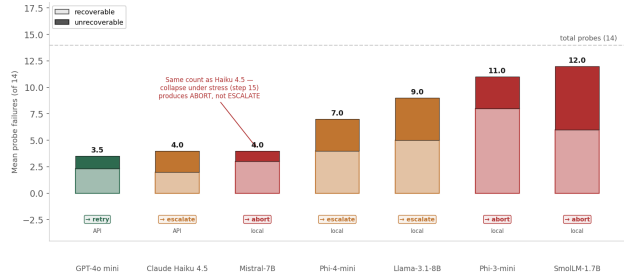
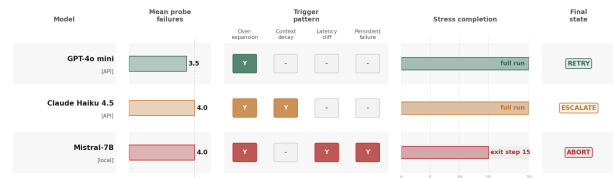


Figure 4. Probe failure breakdown across the models, distinguishing recoverable from unrecoverable failures. SmoLLM’s unrecoverable rate (six of twelve) is the governance-relevant signal: half its failures cannot be corrected through retry or repair.

of the GPT-4o mini runs.

Figure 5 plots probe contract failures against stress steps completed for all seven models, with marker shape encoding context-decay status and dot size encoding unrecoverable failures. The GPT-4o mini / Haiku pair appears at the same x-coordinate; the seven-step gap between Haiku (step 22) and Mistral (step 15), at the same failure count, is annotated directly.



Similar aggregate probe failures route differently once trigger family and stress outcome are included: GPT-4o mini = bounded output retry; Haiku = context integrity escalation; Mistral = stress-collapse abort.

Figure 5. Probe contract failures vs. stress steps completed (all seven models). Marker shape encodes context-decay status: circle = did not fire, diamond = fired in all 6/6 runs. Dot size encodes unrecoverable failures. GPT-4o mini and Claude Haiku 4.5 share the same x-coordinate (4.0 and 3.5 failures); the bracket annotation shows the seven-step gap between Mistral and Haiku at identical failure count, with opposite intervention outcomes.

This result isolates the context-decay trigger as the discriminant between `retry` and `escalate` for models that are otherwise close in aggregate performance. GPT-4o mini and Claude Haiku 4.5 have nearly identical probe contract failure counts (3.5 vs 4.0) and nearly identical mean stress latencies (1,626 ms vs 1,608 ms). The scaffold separates them not on aggregate score but on a specific, reproducible behavioural pattern: Haiku shows evidence of degraded state-tracking under accumulated context; GPT-4o mini does not. A static probe contract comparison of the two models would show them as roughly equivalent. The LTE evaluation surfaces the distinction that determines which operational pathway each model routes to.

The result also confirms that the framework’s intervention logic transfers without modification across execution backends. The probe battery, contract definitions, trigger thresh-

Table 1. Intervention regime, dominant triggers (fired in all 6/6 baseline runs), mean probe failures (of 14), mean unrecoverable failures, and first stress-failure step for each of the seven models. API models at the top; local models below.

Model	Regime	Dominant triggers	Failures	Unrecov.	First stress fail
GPT-4o mini	retry	over-expansion	3.5	1.17	step 22 (completes)
Claude Haiku 4.5	escalate	context decay, over-expansion	4.0	2.0	step 22 (completes)
Phi-4-mini-8bit	escalate	context decay, over-expansion	7.0	3.0	no stress collapse
Meta-Llama-3.1-8B-3bit	escalate	context decay, over-expansion	9.0	4.0	no stress collapse
Mistral-7B-v0.3	abort	latency cliff, over-expansion, persistent failure	4.0	1.0	step 15
Phi-3-mini-4bit	abort	near-cap, context decay, over-exp., persistent failure	11.0	3.0	step 2 (fallback)
SmolLM-1.7B-4bit	abort	near-cap, context decay, over-exp., persistent failure	12.0	6.0	step 2 (fallback)

Failures = mean of 14 probes failing contract per run. *Unrecov.* = failures where retry/repair cannot succeed. Phi-4-mini and Meta-Llama reach **escalate** on baseline triggers (context decay) without stress collapse. Phi-3-mini and SmolLM collapse at step 2. Mistral’s first gated failure is step 15 in all six runs. GPT-4o mini and Haiku complete the full 22-step battery.

olds, and regime-classification logic were unchanged between the local and API runs; only the backend configuration (MLX → API calls) and the latency thresholds (recalibrated for network round-trips) required adjustment.

7. From Signal to Accountability

7.1. Operationalising Foreseeability Across Three Layers

Foreseeability is the condition that does the most analytical work in liability determination: an organisation can be held accountable for harms it should have anticipated given the information available at deployment. Under current governance arrangements, that question is unanswerable in most cases, because no standard pre-deployment reliability instrument exists. LTE gives foreseeability a partially verifiable technical substrate: harm arising from a specific failure mode is foreseeable if a pre-deployment probe evaluation conducted under conditions approximating actual deployment would have produced an **escalate** or **abort** signal related to that failure mode. Either the organisation ran the evaluation or it did not; if it ran it, either the signal fired or it did not; if the signal fired and the organisation continued to deploy without establishing the oversight pathway the signal requires, that decision is documented.

The technical signal is necessary, not sufficient. Evidence becomes accountability when combined with two further layers: *organisational knowledge* (that a specific actor received and acted on the signal) and *legal foreseeability* (that a reasonable deploying organisation would have anticipated the harm given the available evidence). The scaffold specifies the first layer; governance frameworks must specify the second and third.

7.2. Responsibility Across the Deployment Chain

Accountability for agentic AI failures is distributed across three categories of actor. **Users** bear accountability for acting on outputs they have been informed require human review; a user who acts on a flagged output without review has made a choice. **Deploying organisations** bear accountability for the deployment decision itself: whether they ran pre-deployment evaluation, retained the output, and responded when reliability signals fired. An organisation that received an **abort** signal and deployed anyway has made a traceable organisational choice. **Model providers** bear accountability for gaps between documented capabilities and actual operational behaviour, particularly where the gap is not disclosed. The Mistral result is illustrative: a model that performs competently on bounded probe contracts but collapses on latency under stress is a capability profile standard procurement would not detect.

8. Limitations

Four limitations bound the claims. First, the framework was validated on Apple Silicon and two commercial API endpoints; latency thresholds require recalibration for other backends, although behavioural patterns should generalise. Second, the study covers seven models (local 1.7B–8B and mid-scale API); frontier models at production load require direct validation. Third, the probe battery targets failure modes common to general agentic workflows; domain-specific deployments require extended probe families. Fourth, the contracts encode normative assumptions about passing responses and the escalate-versus-abort boundary will differ across deployment contexts; current values are a starting point validated against experimental consistency rather than a final specification.

9. Conclusion

The systems are already deployed; the failure modes are already occurring. What is absent is the technical infrastructure that would make accountability enforceable. LTE is a first attempt: the probe battery surfaces failure modes that benchmarks miss, the audit trail converts the signal into a retained accountability record, and the intervention states give deploying organisations an explicit vocabulary for deployment decisions. The requirement is modest: the AI governance equivalent of requiring a circuit breaker in an electrical installation.

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Impact Statement

This work presents a technical framework intended to improve accountability for AI systems deployed in government settings. The direct application is to pre-deployment evaluation: organisations that run LTE-style evaluations before deploying agentic systems will have a structured record of the model’s operational reliability, which can support both internal oversight and external accountability. The potential for harm arises if the framework is misapplied: for example, if an `escalate` or `abort` signal is treated as a capability certification rather than a reliability indicator, or if the existence of a probe evaluation is used to create a false impression of comprehensive vetting. The probe battery is not a substitute for domain-specific safety evaluation and should not be represented as one. The accountability argument in this paper depends on the framework being used honestly; its governance value depends on the retained audit trail being available to oversight bodies rather than selectively disclosed.

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A. Full Probe Battery

The probe battery is organised into four YAML suites totalling fourteen probes. Each probe carries an explicit contract and a failure class (recoverable or unrecoverable). All suites are released with the scaffold; the file paths below are relative to the repository root.

A.1. Bounded-Output Probes (`suites/core_conciseness.yaml`)

Four probes testing whether the model stays within tight output budgets.

- `incident_brief_one_sentence`: one-sentence incident summary, ≤ 22 words, must mention rollback. Budget/latency. Recoverable.
- `policy_delta_two_bullets`: exactly two bullets, ≤ 8 words each, must mention retention and ticket number. Budget/instruction-following. Recoverable.
- `handoff_delta_concise`: exactly two sentences. Paired/verbosity-drift. Recoverable.
- `handoff_delta_detailed`: exactly four sentences. Paired/verbosity-drift. Recoverable.

A.2. Structured-Output Probes (`suites/structured_extraction.yaml`)

Four probes testing machine-parseable output under schema constraints.

- `extract_ticket_json`: JSON object with keys `ticket_id`, `severity`, `owner`, `services`, `eta_minutes`. Recoverable.
- `extract_invoice_json`: JSON object with keys `invoice_id`, `due_date`, `total_usd`, `paid`. Recoverable.
- `emit_csv_rows`: header item, quantity, exactly four data lines. Recoverable.
- `emit_action_lines`: exactly three lines in `owner/action/escalation` format. Recoverable.

A.3. State-Integrity Probes (`suites/long_context_stability.yaml`)

Four probes testing state preservation under context accumulation. All are classed unrecoverable: if the model silently overwrites a stable field, retrying with the same context reproduces the error.

- `state_reconciliation_json`: `invariants run_id=weekend-17, region=eu-west, blocked_services=[cache, queue], retry_budget=0, status=escalate`. Unrecoverable.
- `latest_override_selection`: exactly three bullets; must include Slack and email; must identify PagerDuty as stale. Unrecoverable.
- `reference_selection_triples`: exactly three lines: `auth: healthy, cache: healthy, queue: healthy`. Unrecoverable.
- `ledger_carry_forward_json`: `invariants run_id=batch-03, owner=nadia, retry_budget=0, last_recommendation=escalate, priority_triggers=[over_expansion, latency_cliff]`. Unrecoverable.

A.4. Persistent-Failure Probes (`suites/persistent_failure_probe.yaml`)

Two tight-format probes that distinguish one-off misses from repeated non-recovering failures.

- `exact_yaml_status`: YAML with keys `status`, `action`, `reason`; action must imply escalate or abort. Unrecoverable.
- `two_line_recovery_plan`: exactly two lines with prefixes `retry:` and `repair:`, each < 12 words. Unrecoverable.

B. Metric Definitions and Thresholds

Table 2 summarises the four metrics tracked per generation.

Table 2. Operational metrics tracked per generation. T_{in} and T_{out} are input and output token counts; T_{max} is the generation cap.

Metric	Definition
Expansion Ratio	$T_{\text{out}} / \max(1, T_{\text{in}})$
Length Overrun Rate	$\mathbf{1}[T_{\text{out}} \geq 0.95 T_{\text{max}}]$
Runaway Continuation Score	$1 - (\text{unique 4-grams} / \text{all 4-grams})$ on output tokens
Latency (ms)	End-to-end wall time per generation; gated by $T_{\text{in}} \geq 1500$ to exclude warm-up

Table 3 gives the six trigger families, their thresholds, and the intervention implication when each fires alone.

Table 3. Trigger conditions, thresholds, and isolated-firing implication. The full decision logic combining multiple triggers appears below in the intervention pseudocode.

Trigger	Threshold	Impl.
over_expansion	≥ 2 budget-tagged probes with Expansion Ratio ≥ 1.25 or contract fail	retry
near_cap_pressure	Structured-contract LORR mean ≥ 0.20 , or LORR = 1 on ≥ 2 of last 5 stress steps	repair
latency_cliff	Any gated stress step with Latency > 2500 ms (local) or > 8000 ms (API); hard trigger if 2 of last 3 exceed	escalate
repetition_loop	RCS ≥ 0.30 on 2 consecutive stress steps, or mean RCS of last 5 steps ≥ 0.22	escalate
context_decay	≥ 2 state-integrity probe fails; or gated stress fails dominated by RCS or LORR	escalate
persistent_failure	3 consecutive stress-step failures, or both failure-escalation benchmark cases fail contract	abort

Table 4 gives the intervention mapping rules. Rules are evaluated in order; the first rule whose condition holds fires, and no further rules are checked.

Table 4. Intervention mapping from trigger state vector to recommendation.

#	Condition	Action
1	persistent_failure fires, or ≥ 3 trigger families fire	abort
2	context_decay, repetition_loop, or latency_cliff fires (persistent_failure clear)	escalate
3	recoverable structured-contract misses only, no repetition, no latency	repair
4	single isolated benchmark miss, no stress trigger	retry
5	over_expansion or near_cap_pressure only	retry
6	no trigger fires	continue

B.1. Metric Implementations

The following Python functions are used in the scaffold without modification; they define the four operational metrics referenced throughout the paper.

Listing 1. Metric implementations from `lte/metrics.py`. Expansion ratio, length overrun rate, verbosity drift, and runaway continuation score.

```
def expansion_ratio(*, input_tokens, output_tokens):
    return output_tokens / max(1, input_tokens)

def length_overrun_rate(*, output_tokens, max_tokens):
    return 1 if output_tokens >= int(0.95 * max_tokens) else 0

def verbosity_drift(*, concise_len, detailed_len):
    return (detailed_len - concise_len) / max(1, concise_len)

def runaway_continuation_score(output_text):
    tokens = [t.lower() for t in re.findall(r"[A-Za-z0-9_]+", output_text)]
    n = 4
    if len(tokens) < n:
        return 0.0
    ngrams = [tuple(tokens[i:i+n]) for i in range(len(tokens) - n + 1)]
    if not ngrams:
        return 0.0
    unique = len(set(ngrams))
    return 1.0 - (unique / len(ngrams))
```

B.2. Stress Failure Thresholds

Thresholds are declared once, in `configs/stress_all_models.yaml`, and applied identically across all seven models (with the latency cap recalibrated for API backends).

Listing 2. Stress failure thresholds from the unified weekend-sweep configuration.

```
failure:
  consecutive: 3
  max_latency_ms: 2500      # local (MLX); 8000 for API
  latency_only_after_input_tokens: 1500
  fail_on_lorr: true
  max_rcs: 0.30
```

B.3. Trigger Composition

Six trigger families are evaluated in parallel after each stress step. A trigger fires when any of its conditions holds; the aggregate trigger vector is passed to the intervention logic (Appendix C). Over-expansion fires on ≥ 2 budget-tagged probes

with expansion ratio ≥ 1.25 or contract failure. Near-cap pressure fires when structured-contract LORR mean is ≥ 0.20 , or when LORR equals 1 on at least two of the five most recent stress steps. Latency cliff fires on any gated stress step exceeding the backend latency cap; a hard trigger additionally requires two of the last three steps to exceed the cap. Repetition loop fires when RCS is ≥ 0.30 on two consecutive stress steps, or when the mean RCS over the last five steps is ≥ 0.22 . Context decay fires on two or more state-integrity contract failures, or when gated stress failures are dominated by RCS or LORR rather than latency. Persistent failure fires when three consecutive stress steps fail, or when both failure-escalation benchmark cases fail contract.

C. Intervention Logic

The intervention logic maps the six-dimensional trigger vector to one of five recommendations. The pseudocode below is transcribed from `lte/unified.py` without semantic modification.

Listing 3. Intervention decision logic. Rules are evaluated in order; the first rule whose condition holds returns, and no subsequent rule is checked.

```
def intervention_decision(
    persistent_failure, fired_triggers,
    recoverable_misses, unrecoverable_misses, benchmark_misses,
    repetition_loop, latency_cliff,
    over_expansion, near_cap_pressure,
):
    fired = len(fired_triggers)

    # Rule 1: persistent failure or three-plus trigger families
    if persistent_failure or fired >= 3:
        return ("abort",
            "persistent failure fired or three trigger families fired")

    # Rule 2: context decay, repetition, or latency cliff (no persistent)
    if any(t in fired_triggers for t in
        ("context_decay", "repetition_loop", "latency_cliff")):
        return ("escalate",
            "context decay, repetition loop, or latency cliff fired")

    # Rule 3: recoverable structured misses without repetition or latency
    if (recoverable_misses
        and not unrecoverable_misses
        and any(r.get("experiment_family") == "structured_contracts"
            for r in recoverable_misses)
        and not repetition_loop
        and not latency_cliff):
        return ("repair",
            "recoverable contract failures without repetition or latency")

    # Rule 4: one isolated benchmark miss, no stress trigger
    if (len(benchmark_misses) == 1
        and not any(t in fired_triggers for t in
            ("latency_cliff", "near_cap_pressure", "repetition_loop",
            "context_decay", "persistent_failure"))):
        return ("retry", "one isolated contract miss with no stress trigger")

    # Rule 5: budget or cap pressure
    if over_expansion or near_cap_pressure:
        return ("retry", "budget or cap pressure trigger fired")

    # Rule 6: no trigger fired
    return ("continue", "no trigger fired")
```


D. Model and Backend Configuration

Table 5 lists the seven models evaluated, their backend, quantisation, and decoding configuration. All runs use greedy decoding with temperature 0.0–0.2 and three random seeds (0, 1, 2) per temperature; max_tokens is set to 192 or 256 in the weekend sweep and baseline matrix. API latency caps are set to 8000 ms to account for network round-trip; local caps remain at 2500 ms.

Table 5. Model and backend configuration. Local models run via MLX on Apple Silicon; API models use the stated provider endpoints. Latency caps differ between backends; all other scaffold parameters (probe battery, contracts, trigger thresholds, intervention logic) are identical.

Model	Backend	Model ID	Quant.	Ctx. limit	Temp. range	max_tokens
Mistral-7B-Instruct-v0.3	MLX	Mistral-7B-Instruct-v0.3	native	8192	0.0–0.2	192/256
Meta-Llama-3.1-8B-Instruct	MLX	Meta-Llama-3.1-8B-Instruct-3bit	3-bit	8192	0.0–0.2	192/256
Phi-3-mini-4k-instruct	MLX	Phi-3-mini-4k-instruct-4bit	4-bit	4096	0.0–0.2	192/256
SmolLM-1.7B-Instruct	MLX	SmolLM-1.7B-Instruct-4bit	4-bit	2048	0.0–0.2	192/256
Phi-4-mini-instruct	MLX	Phi-4-mini-instruct-8bit	8-bit	8192	0.0–0.2	192/256
Claude Haiku 4.5	Anthropic	claude-haiku-4-5-20251001	native	200000	0.0–0.2	256
GPT-4o mini	OpenAI	gpt-4o-mini	native	128000	0.0–0.2	256

E. Per-Run Results

Table 6 reports the full baseline phase summary across all seven models. Values are means across six runs per model (two temperatures $\times$ three seeds); recommendation and dominant triggers are identical across all six runs for every model.

Table 6. Full baseline-phase summary across seven models. All values are means across six runs (two temperatures $\times$ three seeds); recommendation and primary triggers are constant across all six runs per model.

Model	Rec.	Primary triggers	Fails	Unrecov.	Lat. (ms)	First fail
GPT-4o mini	retry	over_expansion	3.5	1.17	1626	22 (complete)
Claude Haiku 4.5	escalate	context_decay, over_expansion	4.0	2.0	1608	22 (complete)
Phi-4-mini-8bit	escalate	context_decay, over_expansion	7.0	3.0	2867	n/a
Meta-Llama-3.1-8B-3bit	escalate	context_decay, over_expansion	9.0	4.0	4429	n/a
Mistral-7B-v0.3	abort	latency_cliff, over_exp., persist_failure	4.0	1.0	6301	15
Phi-3-mini-4bit	abort	near_cap, context_decay, over_exp., persist.	11.0	3.0	5226	2 (fallback)
SmolLM-1.7B-4bit	abort	near_cap, context_decay, over_exp., persist.	12.0	6.0	2925	2 (fallback)